

EXAMPLES USING TWO APPROACHES TO SOLVE $\frac{dy}{dx} + a(x)y = q(x)$

① Solve $\frac{dy}{dx} = 2y - 1$

for the initial condition $y(0) = 0$

Solution with y_p & y_h : $a(x) = 2$
 $q(x) = -1$

Particular solution (y_p):

Let $y_p = A$
 $\frac{dy_p}{dx} = 2y_p - 1$
 $\frac{dA}{dx} = 2A - 1 \Rightarrow A = \frac{1}{2}$
 $y_p = \frac{1}{2}$

Homogenous solution (y_h):

$$\frac{dy_h}{dx} = 2y_h$$

$$\Rightarrow \frac{dy_h}{y_h} = 2dx$$

Integrating: $\int \frac{dy_h}{y_h} = \int 2dx + C$

$$\ln(y_h) = 2x + C \Rightarrow y_h = Ce^{2x}$$

Complete solution: $y = y_h + y_p$

$$\Rightarrow y(x) = \frac{1}{2} + Ce^{2x}$$

Apply initial condition $y(0) = 0$

$$\Rightarrow 0 = \frac{1}{2} + Ce^{2(0)} \Rightarrow C = -\frac{1}{2}$$

$$y(x) = \frac{1}{2} - \frac{1}{2}e^{2x}$$

Solution with $p(x)$:

Integrating factor $p(x)$:

Recall: $p(x) = e^{\int a(x) dx}$

& ODE can be written as $\left[\frac{dy}{dx} + 2y = -1 \right]$

Here: $a(x) = 2$
 $q(x) = -1$

$$\int p(x) dx = \int -2dx + C = -2x + C$$

$$e^{\int a(x) dx} = e^{-2x+C} = Ke^{-2x}$$

$$\Rightarrow p(x) = Ke^{-2x}$$

Multiplying by $p(x)$ on both sides.

ODE: $p(x) \frac{dy}{dx} + p(x) \cdot 2y = -p(x)$

$$\Rightarrow Ke^{-2x} \frac{dy}{dx} + (Ke^{-2x}) \cdot 2y = -Ke^{-2x}$$

Express L.H.S as a derivative of a product.

$$\frac{d}{dx} (e^{-2x} y) = -e^{-2x}$$

Integrate and solve:

$$d(e^{-2x} y) = -e^{-2x} dx$$

$$\int d(e^{-2x} y) = -\int e^{-2x} dx + C$$

$$\Rightarrow e^{-2x} y = -\frac{e^{-2x}}{2} + C$$

$$\therefore y(x) = \frac{1}{2} + Ce^{2x} \text{ Apply } y(0) = 0 \Rightarrow C = -\frac{1}{2}$$

$$y(x) = \frac{1}{2} - \frac{1}{2}e^{2x}$$

- (2) Solve the linear first-order ODE for exponential forcing function (e^{st}) with constant, a :

$$\frac{dy}{dt} = ay + e^{st}$$

Solution with y_p & y_h $a(t) = a$
 $q(t) = e^{st}$

Particular solution (y_p)

Let $y_p = Y \cdot e^{st}$: $\frac{dy_p}{dt} = a y_p + e^{st}$

$$\frac{d(Ye^{st})}{dt} = a(Ye^{st}) + e^{st}$$

$$Y s e^{st} = a Y e^{st} + e^{st}$$

$$Y(s-a) = 1 \Rightarrow Y = \frac{1}{s-a}$$

$$y_p = \frac{e^{st}}{s-a}$$

Homogenous solution (y_h):

$$\frac{dy_h}{dt} = a y_h$$

$$\Rightarrow \frac{dy_h}{y_h} = a dt \leftarrow \text{Integrate}$$

$$\Rightarrow \int \frac{dy_h}{y_h} = \int a dt + C$$

$$\Rightarrow \ln(y_h) = at + C$$

$$\Rightarrow y_h = C e^{at}$$

Complete solution: $y = y_p + y_h$

$$y(t) = \frac{e^{st}}{s-a} + C e^{at}$$



Solution with integrating factor, $p(t)$

Integrating factor $p(t)$:

$$p(t) = e^{\int a(t) dt}$$

for ODE re-arranged as

$$\left[\frac{dy}{dt} - ay = e^{st} \right] \text{ where } a(t) = -a$$

$$q(t) = e^{st}$$

$$\int a(t) dt = \int -a dt + C = -at + C$$

$$p(t) = e^{\int a(t) dt} \Rightarrow p(t) = K e^{-at}$$

Multiplying on both sides of ODE by

$$p(t): p(t) \cdot \frac{dy}{dt} - p(t)ay = p(t)e^{st}$$

$$\Rightarrow K e^{-at} \frac{dy}{dt} - (K e^{-at}) ay = K e^{-at} e^{st}$$

$$\Rightarrow e^{-at} \frac{dy}{dt} - e^{-at} ay = e^{t(s-a)}$$

Express the L.H.S. as a product of a derivative:

$$\Rightarrow \frac{d(e^{-at} y)}{dt} = e^{t(s-a)}$$

Integrate and solve:

$$\int d(e^{-at} y) = \int e^{t(s-a)} dt + C$$

$$\Rightarrow e^{-at} y = \frac{e^{st} e^{-at}}{s-a} + C$$

$$\Rightarrow y(t) = \left(\frac{e^{st}}{s-a} \right) + C e^{at}$$

Note: The solutions are presented without considering "resonance" (Case where $s-a=0$)

(3) R.V.B.

(3) Solve $y' + y = 2$ for $[y(0) = 0]$ Solution with y_p & y_h :Rearranging the ODE $y' = -y + 2$ for the form $\{y' = a(x)y + q(x)\}$ where $\begin{cases} a(x) = -1 \\ q(x) = 2 \end{cases}$ Particular solution (y_p):Let $y_p = A$; $\frac{dy_p}{dx} = -y_p + 2$

$$\frac{dA}{dx} = 0 \Rightarrow A + 2 \Rightarrow A = -2 \Rightarrow y_p = 2$$

Homogenous solution (y_h):

$$\frac{dy_h}{dx} = -y_h \Rightarrow \frac{dy_h}{y_h} = -1 \cdot x$$

Integrating

$$\int \frac{dy_h}{y_h} = \int -dx + C$$

$$\Rightarrow \ln(y_h) = -x + C \Rightarrow y_h = Ce^{-x}$$

Complete solution: $y = y_p + y_h$

$$y(x) = 2 + Ce^{-x}$$

$$\text{Apply } y(0) = 0 \Rightarrow 0 = 2 + Ce^0$$

$$\Rightarrow C = -2$$

$$y(x) = 2(1 - e^{-x})$$

Do MATH
It's COOL!Solution with integrating factor $f(x)$:

$$y' + y = 2$$

is of the form $\{y' = a(x)y + q(x)\}$

$$\text{where } \begin{cases} a(x) = 1 \\ q(x) = 2 \end{cases}$$

Integrating factor: $f(x)$:

$$f(x) = e^{\int a(x) dx}$$

$$\int a(x) dx = \int 1 dx + C = x + C$$

$$f(x) = e^{\int a(x) dx} = e^{x+C} \Rightarrow f(x) = Ke^x$$

Multiplying both sides of ODE by $f(x)$:

$$f(x)y' + f(x)y = 2f(x)$$

$$\Rightarrow Ke^x \frac{dy}{dx} + Ke^x y = 2Ke^x$$

$$\Rightarrow e^x \frac{dy}{dx} + e^x y = 2e^x$$

Express the L.H.S as derivative of a product

$$\frac{d(e^x y)}{dx} = 2e^x$$

Integrate and solve:

$$d(e^x y) = 2e^x dx$$

$$\int d(e^x y) = \int 2e^x dx + C$$

$$\Rightarrow e^x y = 2e^x + C$$

$$\Rightarrow y(x) = 2 + Ce^{-x}$$

$$\text{Apply } y(0) = 0 \Rightarrow 0 = 2 + Ce^0$$

$$\Rightarrow C = -2$$

$$y(x) = 2(1 - e^{-x})$$

(4) KVB

④ Solve. $\frac{dy}{dx} = y + \frac{11}{8}e^{-x/3}$ for $y(0) = -1$

Solution with y_p & y_h :

Given ODE: $\frac{dy}{dx} = y + \frac{11}{8}e^{-x/3}$

is of the form $\left\{ \frac{dy}{dx} = a(x)y + q(x) \right\}$

Where $\left\{ \begin{aligned} a(x) &= 1 \\ q(x) &= \frac{11}{8}e^{-x/3} \end{aligned} \right\}$

Particular solution (y_p):

Let $y_p = Ye^{-x/3}$; $\frac{dy_p}{dx} = y_p + \frac{11}{8}e^{-x/3}$

$\frac{d}{dx}(Ye^{-x/3}) = Ye^{-x/3} + \frac{11}{8}e^{-x/3}$

$Y(-\frac{1}{3})e^{-x/3} = Ye^{-x/3} + \frac{11}{8}e^{-x/3}$

$\Rightarrow Y(-\frac{1}{3} - 1) = \frac{11}{8} \Rightarrow Y(-\frac{4}{3}) = \frac{11}{8}$

$\Rightarrow Y = \frac{-33}{32} \Rightarrow y_p = \left(\frac{-33}{32}\right)e^{-x/3}$

Homogenous solution (y_h):

$\frac{dy_h}{dx} = y_h$

$\Rightarrow \frac{dy_h}{y_h} = dx$ Integrating

$\Rightarrow \int \frac{dy_h}{y_h} = \int dx + C \Rightarrow \ln y_h = x + C$

$\Rightarrow y_h = Ce^x$

Solution with integrating factor $\rho(x)$:

Rearranging the given ODE

$\frac{dy}{dx} - y = \frac{11}{8}e^{-x/3}$; It is of the

form $\left\{ \frac{dy}{dx} + a(x)y = q(x) \right\}$

Where $\left\{ \begin{aligned} a(x) &= -1 \\ q(x) &= \frac{11}{8}e^{-x/3} \end{aligned} \right\}$

Integrating factor $\rho(x)$:

$\rho(x) = e^{\int a(x)dx}$

$\int a(x)dx = \int -1dx = -x + C$

$\rho(x) = e^{\int a(x)dx} = e^{-x+C}$

$\Rightarrow \rho(x) = Ke^{-x}$

Multiplying on both sides of

the ODE by $\rho(x)$:

$\rho(x) \cdot \frac{dy}{dx} - \rho(x) \cdot y = \rho(x) \cdot \frac{11}{8}e^{-x/3}$

$\Rightarrow \cancel{K}e^{-x} \frac{dy}{dx} - \cancel{K}e^{-x}y = (\cancel{K}e^{-x}) \cdot \frac{11}{8}e^{-x/3}$

$\Rightarrow e^{-x} \frac{dy}{dx} - e^{-x}y = e^{-x} \left(\frac{11}{8}\right)e^{-x/3}$

Express the L.H.S of ODE as a derivative of a product:

$\frac{d(e^{-x}y)}{dx} = \left(\frac{11}{8}\right)e^{-x-x/3}$

$\Rightarrow \frac{d(e^{-x}y)}{dx} = \frac{11}{8}e^{-4x/3}$

Solution with y_p & y_h
continued

Complete solution:

$$y = y_p + y_h$$

$$y(x) = -\frac{33}{32} e^{-x/3} + C e^x$$

Apply $y(0) = -1$

$$-1 = -\frac{33}{32} e^{-0/3} + C e^{0/3}$$

$$\Rightarrow C = \frac{33}{32} - 1 \Rightarrow \boxed{C = \frac{1}{32}}$$

$$y(x) = -\frac{33}{32} e^{-x/3} + \frac{1}{32} e^x$$

$$\boxed{y(x) = \frac{1}{32} (e^x - 33 e^{-x/3})}$$



Solution with integrating factor $g(x)$; continued

$$\Rightarrow d(e^{-x} y) = \frac{11}{8} e^{-4x/3} dx$$

Integrate and solve:

$$\int d(e^{-x} y) = \int \frac{11}{8} e^{-4x/3} dx + C$$

$$\Rightarrow e^{-x} y = \frac{11}{8} \left[\frac{e^{-4x/3}}{(-4/3)} \right] + C$$

$$= \left(-\frac{33}{32} \right) e^{-4x/3} + C$$

$$\Rightarrow y(x) = \left(-\frac{33}{32} \right) \frac{e^{-4x/3}}{e^{-x}} + \frac{C}{e^{-x}}$$

Apply $y(0) = -1$

$$\Rightarrow -1 = \left(-\frac{33}{32} \right) \frac{e^{-4 \cdot 0/3}}{e^{-0}} + \frac{C}{e^{-0}}$$

$$\Rightarrow C = \frac{33}{32} - 1 \Rightarrow \boxed{C = \frac{1}{32}}$$

$$\therefore y(x) = -\frac{33}{32} e^{-4x/3+x} + \frac{1}{32} e^x$$

$$y(x) = -\frac{33}{32} e^{-x/3} + \frac{1}{32} e^x$$

$$\boxed{y(x) = \frac{1}{32} (e^x - 33 e^{-x/3})}$$

(5)

Solve

$$y' - 2xy = e^{x^2}$$

(6)

KVB

Solution with y_p & y_h :

Rearranging the given ODE

$\{y' = 2xy + e^{x^2}\}$ where

$$\begin{cases} a(x) = 2x \\ q(x) = e^{x^2} \end{cases}$$

Particular solution (y_p):

Let $y_p = Ax e^{x^2}$

$$\frac{dy_p}{dx} = 2x y_p + e^{x^2}$$

$$\Rightarrow \frac{d}{dx}(Ax e^{x^2}) = (2x)(Ax e^{x^2}) + e^{x^2}$$

$$\Rightarrow (Ax)(2x)e^{x^2} + A e^{x^2} \frac{dx}{dx} = (2x)(Ax e^{x^2}) + e^{x^2}$$

$$\Rightarrow \cancel{2Ax^2 e^{x^2}} + A e^{x^2} = \cancel{2Ax^2 e^{x^2}} + e^{x^2}$$

$$\Rightarrow \boxed{A=1} \Rightarrow \boxed{y_p = x e^{x^2}}$$

Homogenous solution (y_h):

$$\frac{dy_h}{dx} = 2x y_h$$

$$\Rightarrow \frac{dy_h}{y_h} = 2x dx \leftarrow \text{Integrating}$$

$$\Rightarrow \int \frac{dy_h}{y_h} = \int 2x dx$$

$$\Rightarrow \ln(y_h) = \frac{2x^2}{2} + C$$

Solution with integrating factor $f(x)$:

Given ODE $y' - 2xy = e^{x^2}$ is of the form: $\{y' + a(x)y = q(x)\}$

$$\text{where } \begin{cases} a(x) = -2x \\ q(x) = e^{x^2} \end{cases}$$

Integrating factor $f(x)$:

$$f(x) = e^{\int a(x) dx}$$

$$\int a(x) dx = \int -2x dx = -\frac{2x^2}{2} + C$$

$$f(x) = e^{\int a(x) dx} = e^{-x^2 + C}$$

$$\Rightarrow \boxed{f(x) = K e^{-x^2}}$$

Multiplying both sides of the ODE by $f(x)$:

$$f(x) \frac{dy}{dx} - f(x) 2xy = f(x) e^{x^2}$$

$$\Rightarrow \cancel{K e^{-x^2} \frac{dy}{dx}} - \cancel{K e^{-x^2} (2xy)} = \cancel{K e^{-x^2} e^{x^2}}$$

$$\Rightarrow e^{-x^2} \frac{dy}{dx} - e^{-x^2} (2xy) = 1$$

Express the L.H.S as the derivative of a product:

$$\frac{d}{dx}(e^{-x^2} y) = 1$$

Integrate and solve:

$$d(e^{-x^2} y) = 1 dx$$

Solution with $y_p \neq y_h$
continued:

$$y_h = Ce^{x^2}$$

Complete solution: $y = y_p + y_h$

$$y(x) = xe^{x^2} + Ce^{x^2}$$

$$y(x) = e^{x^2} [x + C]$$

Solution with Integrating
factor $\rho(x)$ continued:

$$\int d(e^{-x^2} y) = \int 1 dx + C$$

$$\Rightarrow e^{-x^2} y = x + C$$

$$\Rightarrow y(x) = e^{x^2} [x + C]$$



Some notes:

- ① The guess for the particular solution was not obvious.
- ② There is a trick one can use if the y_p guess is not easy enough to make.
- ③ Look for the integrating factor $\rho(x)$ and use the formula below:

$$y_p(x) = \frac{1}{\rho(x)} \int \rho(x) q(x) dx \quad \left\{ \begin{array}{l} \text{where } \rho(x) = e^{\int a(x) dx} \\ \rho(x) = e^{x^2} \end{array} \right.$$

In this example: CAVEAT (NONE OF THIS IS AN EXACT SCIENCE)

- ① Note that the ODE was not rearranged to in the integrating factor - solution. It was already in the "correct" form: $[y' + a(x)y = q(x)] \Leftrightarrow y' - 2xy = e^{x^2}$
- ② Use these $a(x) \neq q(x)$ functions where $\begin{cases} a(x) = -2 \\ q(x) = e^{x^2} \end{cases}$

to get, $\rho(x) = ke^{-x^2}$

$$\text{Now: } y_p = \frac{1}{ke^{-x^2}} \int ke^{-x^2} e^{x^2} dx \Rightarrow y_p = Ax e^{x^2}$$

(6)

Solve

$$2xy' + y = 10\sqrt{x}$$

(8)

KVB

Solution with y_p & y_h :

Rearranging the given ODE,

$$\left\{ \begin{aligned} y' &= -\frac{y}{2x} + \frac{5\sqrt{x}}{2x} \\ \Rightarrow y' &= -\frac{y}{2x} + \frac{5\sqrt{x}}{x} \end{aligned} \right.$$

$$\text{where } \{a(x) = -\frac{1}{2x}; q(x) = \frac{5\sqrt{x}}{x}\}$$

Particular solution (y_p):

Note: y_p was not something I could guess. So I chose to use $P(x)$.

$$y_p = \frac{1}{P(x)} \int P(x) q(x) dx$$

$$= \frac{1}{Kx^{1/2}} \int Kx^{1/2} \frac{5\sqrt{x}}{x} dx$$

$$= \frac{1}{Kx^{1/2}} \int K \cdot 5 \frac{x}{x} dx$$

$$= \frac{K}{Kx^{1/2}} \int 5 dx = \frac{5x}{\sqrt{x}}$$

$$= \frac{5x\sqrt{x}}{\sqrt{x}\sqrt{x}} = \frac{5x\sqrt{x}}{x}$$

$$\therefore y_p = A\sqrt{x} \leftarrow \text{"guess"}$$

Homogenous solution (y_h):

$$\frac{dy_h}{dx} = -\frac{y_h}{2x}$$

$$\frac{dy_h}{y_h} = -\frac{1}{2x} dx \leftarrow \text{Integrating}$$

Solution with integrating factor, $P(x)$:

Rearranging the given ODE;

$$\left\{ \begin{aligned} y' + \frac{y}{2x} &= \frac{5\sqrt{x}}{2x} \\ \Rightarrow y' + \frac{y}{2x} &= \frac{5\sqrt{x}}{x} \end{aligned} \right.$$

$$\text{where } \{a(x) = \frac{1}{2x}; q(x) = \frac{5\sqrt{x}}{x}\}$$

Integrating factor, $P(x)$:

$$P(x) = e^{\int a(x) dx}$$

$$\int a(x) dx = \int \frac{1}{2x} dx = \frac{1}{2} \ln(x) + C$$

$$= \ln x^{1/2} + \ln(C)$$

$$= \ln(Kx^{1/2})$$

$$e^{\int a(x) dx} = \exp[\ln(Kx^{1/2})]$$

$$\rightarrow P(x) = Kx^{1/2}$$

Multiplying on both sides by $P(x)$:

$$P(x) \frac{dy}{dx} + P(x) \frac{y}{2x} = P(x) \frac{5\sqrt{x}}{x}$$

$$\Rightarrow Kx^{1/2} \frac{dy}{dx} + Kx^{1/2} \frac{y}{2x} = Kx^{1/2} \frac{5\sqrt{x}}{x}$$

$$\Rightarrow x^{1/2} \frac{dy}{dx} + \frac{y}{2x^{-1/2}} = \frac{5x}{x}$$

Express the L.H.S as the derivative of a product:

$$\frac{d}{dx}(x^{1/2}y) = 5$$

SOLUTION WITH $y_p \in y_h$ continued:

$$\int \frac{dy_h}{y_h} = \int \frac{-1}{2x} dx + K$$

$$\ln(y_h) = -\frac{1}{2} \ln(x) + K$$

$$= \ln(x^{-1/2}) + \ln(C)$$

$$\ln(y_h) = \ln(C x^{-1/2})$$

$$\Rightarrow y_h = \exp[\ln(C x^{-1/2})]$$

$$y_h = C x^{-1/2}$$

Particular solution (y_p):

$$y_p = A \sqrt{x}$$

$$\frac{d y_p}{d x} = \frac{-y_p}{2x} + \frac{5\sqrt{x}}{x}$$

$$\frac{d(A\sqrt{x})}{d x} = \frac{-A\sqrt{x}}{2x} + \frac{5\sqrt{x}}{x}$$

$$\Rightarrow A \frac{d\sqrt{x}}{d x} = -\frac{A}{2} \frac{\sqrt{x}}{x} \frac{\sqrt{x}}{\sqrt{x}} + 5 \frac{\sqrt{x}}{x} \frac{\sqrt{x}}{\sqrt{x}}$$

$$\Rightarrow \frac{A}{2} \cdot \frac{1}{\sqrt{x}} = -\frac{A}{2} \cdot \frac{1}{\sqrt{x}} + 5 \cdot \frac{1}{\sqrt{x}}$$

$$\Rightarrow \frac{A}{2} \left(\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{x}} \right) = 5 \cdot \frac{1}{\sqrt{x}}$$

$$\Rightarrow \frac{2 \cdot A}{2} \left(\frac{1}{\sqrt{x}} \right) = 5 \cdot \frac{1}{\sqrt{x}} \Rightarrow A = 5$$

$$y_p = 5\sqrt{x}$$

Complete solution: $y = y_p + y_h$

$$y(x) = 5\sqrt{x} + C x^{-1/2}$$

$$\therefore y(x) = 5\sqrt{x} + \frac{C}{\sqrt{x}}$$

Solution with integrating factor continued:

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KVB

$$d(x^{1/2} y) = 5 dx$$

Integrate and solve:

$$\int d(x^{1/2} y) = \int 5 dx + C$$

$$x^{1/2} y = 5x + C$$

$$\therefore y(x) = \frac{5x}{x^{1/2}} + \frac{C}{x^{1/2}}$$

$$y(x) = 5\sqrt{x} + \frac{C}{\sqrt{x}}$$

(7)

Solve

$$x \cdot y' + 5y = 7x^2$$

for $[y(2) = 5]$ Solution with $y_p \neq y_h$:

Rearranging the ODE:

$$y' = -\frac{5y}{x} + \frac{7x^2}{x}$$

$$\Rightarrow y' = -\frac{5y}{x} + 7x \quad \left\{ \begin{array}{l} \text{where } a(x) = -\frac{5}{x} \\ q(x) = 7x \end{array} \right.$$

Particular solution (y_p):

Note: I will use $p(x)$ i.e. the integrating factor to formulate y_p .

$$y_p \approx \frac{1}{p(x)} \cdot \int p(x) q(x) dx$$

$$y_p = \frac{1}{Kx^5} \int Kx^5 \cdot 7x dx$$

$$= \frac{7K}{x^5} \int x^6 dx$$

$$= \frac{7K}{x^5} \cdot \frac{x^7}{7} = Kx^2$$

$$\Rightarrow y_p = Kx^2 \rightarrow \text{reasonable guess}$$

Now: $\frac{dy_p}{dx} = -\frac{5y_p}{x} + 7x$

$$\Rightarrow \frac{d(Kx^2)}{dx} = -\frac{5Kx^2}{x} + 7x$$

Solution with integrating factor, $p(x)$:

Rearranging the ODE:

$$y' + \frac{5y}{x} = \frac{7x^2}{x}$$

$$\Rightarrow y' + \frac{5y}{x} = 7x \quad \left\{ \begin{array}{l} \text{where } a(x) = \frac{5}{x} \\ q(x) = 7x \end{array} \right.$$

Integrating factor, $p(x)$:

$$p(x) = e^{\int a(x) dx}$$

$$\int a(x) dx = \int \frac{5}{x} dx + C$$

$$= 5 \ln(x) + C$$

$$= \ln(x^5) + \ln(C)$$

$$= \ln(Kx^5)$$

$$e^{\int a(x) dx} = \exp[\ln(Kx^5)]$$

$$\therefore p(x) = Kx^5$$

Multiplying both sides by $p(x)$:

$$p(x) \frac{dy}{dx} + p(x) \frac{5y}{x} = p(x) 7x$$

$$\Rightarrow Kx^5 \frac{dy}{dx} + Kx^5 \frac{5y}{x} = Kx^5 \cdot 7x$$

$$\Rightarrow x^5 \frac{dy}{dx} + x^4 5y = 7x^6$$

Express L.H.S as a derivative of a product:

Solution with $y_p \in y_h$
continued:

$$K \frac{dx^2}{dx} = - \frac{5Kx^7}{x} + 7x$$

$$2Kx = -5Kx + 7x$$

$$7Kx = 7x \Rightarrow \boxed{K=1}$$

$$\Rightarrow \boxed{y_p = x^2}$$

Homogenous solution (y_h):

$$\frac{dy_h}{dx} = - \frac{5}{x} y_h$$

$$\Rightarrow \frac{dy_h}{y_h} = - \frac{5}{x} dx$$

Integrate

$$\Rightarrow \int \frac{dy_h}{y_h} = - \int \frac{5}{x} dx + K$$

$$\Rightarrow \ln(y_h) = -5 \ln(x) + K$$

$$\Rightarrow \ln(y_h) = \ln(x^{-5}) + \ln(K)$$

$$\Rightarrow \ln(y_h) = \ln(Kx^{-5})$$

$$\Rightarrow y_h = \exp[\ln(Kx^{-5})]$$

$$\Rightarrow \boxed{y_h = Cx^{-5}}$$

Complete solution: $\boxed{y = y_p + y_h}$

$$y(x) = x^2 + Cx^{-5}$$

Apply $y(2)=5$: $5 = 2^2 + C \cdot 2^{-5}$

$$\Rightarrow \boxed{C = \frac{5-4}{32^{-1}} = 32}$$

$$\Rightarrow \boxed{y(x) = x^2 + \frac{32}{x^5}}$$

Solution with integrating factor continued:

$$\frac{d(x^5 y)}{dx} = 7x^6$$

$$\Rightarrow d(x^5 y) = 7x^6 dx$$

Integrate and solve:

$$\int d(x^5 y) = \int 7x^6 dx + C$$

$$x^5 y = \frac{7x^7}{7} + C$$

$$y = \frac{x^7}{x^5} + \frac{C}{x^5}$$

$$\therefore y(x) = x^2 + \frac{C}{x^5}$$

Apply $y(2)=5$:

$$y(2)=5 = 2^2 + \frac{C}{2^5}$$

$$\Rightarrow C = 32(5-4) = 32$$

$$\therefore \boxed{C=32}$$

$$\therefore \boxed{y(x) = x^2 + \frac{32}{x^5}}$$

⑧

Solve

$$(x^2+1) \frac{dy}{dx} + 3xy = 6x$$

Solution with y_p & y_h :

Rearranging the given ODE:

$$\frac{dy}{dx} + \frac{3xy}{x^2+1} = \frac{6x}{x^2+1}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{3xy}{x^2+1} + \frac{6x}{x^2+1}$$

$$\text{where } a(x) = -\frac{3x}{x^2+1}; \quad q(x) = \frac{6x}{x^2+1}$$

Particular solution (y_p):Note: There are two way (at least!) to guess y_p .① I recognize that $q(x) = 2a(x)$

$$\text{i.e., } \frac{6x}{x^2+1} = 2 \cdot \left(\frac{3x}{x^2+1} \right) = q$$

$$\Rightarrow q(x) = 2a(x)$$

So 4 reasonable guess would

$$\text{by } \boxed{y_p = 2A}$$

② Use the $f(x)$ we found from the integrating factor

$$\boxed{y_p = \frac{A}{f(x)} \int f(x) q(x) dx}$$

$$\Rightarrow y_p = \frac{A}{K(x^2+1)^{3/2}} \int K(x^2+1)^{3/2} \cdot \frac{6x}{x^2+1} dx$$

$$= \frac{AK}{K(x^2+1)^{3/2}} \int 6x(x^2+1)^{1/2} dx$$

Solution with integrating factor, $f(x)$:

Rearranging the given ODE:

$$\frac{dy}{dx} + \frac{3xy}{(x^2+1)} = \left(\frac{6x}{x^2+1} \right)$$

$$\text{where } \begin{cases} a(x) = \left(\frac{3x}{x^2+1} \right) \\ q(x) = \left(\frac{6x}{x^2+1} \right) \end{cases}$$

Integrating factor $f(x)$:

$$f(x) = e^{\int a(x) dx}$$

$$\int a(x) dx = \int \left(\frac{3x}{x^2+1} \right) dx + C$$

$$\text{Let } x^2+1 = u \Rightarrow du = 2x dx$$

$$\therefore x dx = \frac{du}{2}$$

$$\Rightarrow \int \frac{3x}{x^2+1} dx + C = \int \frac{3}{2} \frac{du}{u} + C$$

$$\Rightarrow \frac{3}{2} \ln u + C = \frac{3}{2} \ln(u) + \ln(C)$$

$$\Rightarrow \int \frac{3x}{x^2+1} dx + C = \frac{3}{2} \ln(x^2+1) + \ln(C)$$

$$= \ln(x^2+1)^{3/2} + \ln(C)$$

$$= \ln[K(x^2+1)^{3/2}]$$

$$\therefore f(x) = \exp[\ln\{K(x^2+1)^{3/2}\}]$$

$$\boxed{f(x) = K(x^2+1)^{3/2}}$$

Solution with y_p & y_h
Continued:

$$y_p = \frac{A}{(x^2+1)^{3/2}} \int 6x(x^2+1)^{1/2} dx$$

$$\text{Let } u = (x^2+1) \Rightarrow du = 2x dx \\ \Rightarrow \frac{du}{2} = x dx$$

$$y_p = \frac{A}{u^{3/2}} \int 6u^{1/2} \frac{du}{2} \\ = \frac{3A}{u^{3/2}} \int u^{1/2} du \\ = \left(\frac{3A}{u^{3/2}} \right) \left(\frac{u^{3/2}}{3/2} \right) \\ = \left(\frac{3A}{u^{3/2}} \right) \left(\frac{2}{3} \right) (u^{3/2})$$

$$\Rightarrow \boxed{y_p = 2A}$$

$$\therefore \frac{dy_p}{dx} = - \left(\frac{3x y_p}{x^2+1} \right) + \left(\frac{6x}{x^2+1} \right)$$

$$\Rightarrow \frac{d(2A)}{dx} = - \left(\frac{3x(2A)}{x^2+1} \right) + \left(\frac{6x}{x^2+1} \right)$$

$$\Rightarrow \left(\frac{6x}{x^2+1} \right) A = \left(\frac{6x}{x^2+1} \right) \Rightarrow \boxed{A=1}$$

$$\therefore \boxed{y_p = 2}$$

Homogenous solution (y_h):

$$\frac{dy_h}{dx} = - \frac{3x y_h}{x^2+1}$$

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Solution with
integrating factor continued:

Multiplying both sides by
 $S(x)$:

$$P(x) \frac{dy}{dx} + P(x) \left(\frac{3xy}{x^2+1} \right) = P(x) \left(\frac{6x}{x^2+1} \right)$$

$$\Rightarrow \cancel{K}(x^2+1)^{3/2} \frac{dy}{dx} + \cancel{K}(x^2+1)^{3/2} \left(\frac{3xy}{x^2+1} \right) \\ = \cancel{K}(x^2+1)^{3/2} \left(\frac{6x}{x^2+1} \right)$$

$$\Rightarrow (x^2+1)^{3/2} \frac{dy}{dx} + (x^2+1)^{1/2} (3xy)$$

$$= (x^2+1)^{3/2} \left(\frac{6x}{x^2+1} \right)$$

Express the L.H.S. as the derivative of a product!

$$\Rightarrow \frac{d}{dx} [(x^2+1)^{3/2} y] = (x^2+1)^{3/2} \left(\frac{6x}{x^2+1} \right)$$

$$\Rightarrow \frac{d}{dx} [(x^2+1)^{3/2} y] = (6x)(x^2+1)^{1/2}$$

Integrate and solve:

$$\int d[(x^2+1)^{3/2} y] = \int 6x(x^2+1)^{1/2} dx + C$$

$$\Rightarrow (x^2+1)^{3/2} y = \int 6x(x^2+1)^{1/2} dx + C$$

On the R.H.S.:

$$\text{Let } u = (x^2+1) \Rightarrow du = 2x dx \\ \Rightarrow \frac{du}{2} = x dx$$

$$\therefore \int 6x(x^2+1)^{1/2} dx = \int 6u^{1/2} \frac{du}{2} + C \\ = 3 \int u^{1/2} du + C$$

Solution with y_p & y_h
continued:

$$\frac{dy_h}{y_h} = -\left(\frac{3x}{x^2+1}\right) dx$$

Integrating

$$\int \frac{dy_h}{y_h} = - \int \left(\frac{3x}{x^2+1}\right) dx + C$$

On R.H.S.: $\xrightarrow{\quad}$

$$\text{Let } u = x^2 + 1 \Rightarrow du = 2x dx$$

$$\Rightarrow x dx = \frac{du}{2}$$

$$\therefore - \int \frac{3x dx}{x^2+1} = - \int \frac{3}{2} \frac{du}{u}$$

$$= -\frac{3}{2} \ln(u)$$

$$= \ln(u^{-3/2})$$

$$= \ln(x^2+1)^{-3/2}$$

$$\Rightarrow \ln(y_h) = \ln(x^2+1)^{-3/2} + \ln(C)$$

$$= \ln[C(x^2+1)^{-3/2}]$$

$$\Rightarrow y_h = \exp[\ln\{C(x^2+1)^{-3/2}\}]$$

$$\boxed{y_h = C(x^2+1)^{-3/2}}$$

Complete solution: $y = y_p + y_h$

$$y(x) = 2 + C(x^2+1)^{-3/2}$$

$$\therefore \boxed{y(x) = 2 + \frac{C}{(x^2+1)^{3/2}}}$$

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Solution with integrating
factor, $f(x)$ continued:

$$\int 6x(x^2+1)^{1/2} dx = 3 \int u^{1/2} du + C$$

$$= 3 \cdot \frac{u^{3/2}}{3/2} + C$$

$$= \cancel{3} \left(\frac{2}{\cancel{3}}\right) u^{3/2} + C$$

$$= 2(x^2+1)^{3/2} + C$$

$$\therefore (x^2+1)^{3/2} y = 2(x^2+1)^{3/2} + C$$

$$\Rightarrow \boxed{y(x) = 2 + \frac{C}{(x^2+1)^{3/2}}}$$

